

Algebra: Log $\leftrightarrow$ Exponential Conversion

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DIRECTIONS

Convert between logarithmic form $\log_b(x)=y$ and exponential form $b^y=x$.

1 Convert to logarithmic form:

$$4^3 = 64$$

Answer: _____

2 Convert to exponential form:

$$\log_3 27 = 3$$

Answer: _____

3 Convert to logarithmic form:

$$5^{-2} = \frac{1}{25}$$

Answer: _____

4 Convert to logarithmic form:

$$2^5 = 32$$

Answer: _____

5 Convert to logarithmic form:

$$3^4 = 81$$

Answer: _____

6 Convert to exponential form:

$$\log_2 8 = 3$$

Answer: _____

7 Convert to exponential form:

$$\log_5 125 = 3$$

Answer: _____

8 Convert to exponential form:

$$\log_4 16 = 2$$

Answer: _____

9 Convert to exponential form:

$$\log_7 49 = 2$$

Answer: _____

10 Convert to logarithmic form:

$$10^3 = 1000$$

Answer: _____



Log/Exponential Conversion Worksheet

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Answer Key & Solutions

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TEACHER NOTES These two forms are inverses of each other. Practice going both directions fluently.

1 Convert to logarithmic form:

$$= \log_4 64 = 3$$

$$4^3 = 64$$

2 Convert to exponential form:

$$= 3^3 = 27$$

$$\log_3 27 = 3$$

3 Convert to logarithmic form:

$$= \log_5 \frac{1}{25} = -2$$

$$5^{-2} = \frac{1}{25}$$

4 Convert to logarithmic form:

$$= \log_2 32 = 5$$

$$2^5 = 32$$

5 Convert to logarithmic form:

$$= \log_3 81 = 4$$

$$3^4 = 81$$

6 Convert to exponential form:

$$= 2^3 = 8$$

$$\log_2 8 = 3$$

7 Convert to exponential form:

$$= 5^3 = 125$$

$$\log_5 125 = 3$$

8 Convert to exponential form:

$$= 4^2 = 16$$

$$\log_4 16 = 2$$

9 Convert to exponential form:

$$= 7^2 = 49$$

$$\log_7 49 = 2$$

10 Convert to logarithmic form:

$$= \log_{10} 1000 = 3$$

$$10^3 = 1000$$



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